

Thesis Proposal

Topic: An Investigation into Multimodal Emotion Recognition Using Facial Expressions and Speech Signals Using Deep Learning Methods.

Abstract

Emotion recognition enhances human-computer interaction by enabling machines to understand human feelings. This research proposes a multimodal deep learning system combining facial expressions and speech signals to improve accuracy and robustness. The system integrates CNN-based facial analysis and spectrogram-based speech analysis with a fusion mechanism for improved prediction.

Introduction

Human emotions are expressed through multiple modalities such as facial expressions and speech. Most systems rely on a single modality, limiting performance. This research focuses on combining both modalities for better results.

Problem Statement

Existing systems rely on single modalities and perform poorly in real-world conditions. The goal is to build a robust multimodal system for accurate emotion recognition.

Research Gap

1. Over-reliance on unimodal systems: Many existing approaches focus on either facial expressions or speech alone, limiting robustness since human emotions are inherently multimodal.
2. Inefficient fusion techniques: Current multimodal models often use simple or poorly optimized fusion strategies, failing to fully capture the complex relationships between visual and audio cues.
3. Poor real-world performance: Models trained on controlled datasets struggle in real-world conditions due to noise, lighting variations, and diverse user behaviors.
4. Lack of real-time implementation: Many proposed systems are computationally expensive and not suitable for real-time emotion recognition applications.
6. Limited use of advanced architecture: Recent deep learning methods for instance, transformers and hybrid models are still underutilized in combining facial and speech modalities effectively.

Objectives

1. To develop a multimodal emotion recognition model that integrates both facial expressions and speech signals.
2. To build individual models for facial expression analysis and speech emotion recognition.
3. To design an effective fusion strategy for combining visual and audio features.

- 4.To evaluate the performance of the proposed system using standard metrics such as accuracy, precision, recall, and F1-score.
- 5.To deploy a real-time emotion recognition system suitable for practical applications

Methodology

1. Data Collection and Preprocessing: Utilizing publicly available datasets for facial expressions and speech signals. Perform preprocessing such as face detection, image normalization, and noise reduction in audio.
2. Facial Emotion Model: Develop a Convolutional Neural Network (CNN) to extract and learn discriminative features from facial images.
3. Speech Emotion Model: Convert speech signals into spectrograms and use deep learning models to extract relevant emotional features.
4. Feature Fusion: Combine features from both modalities using an appropriate fusion strategy like early, late, or hybrid fusion.
5. Training and Evaluation: Train the multimodal system and evaluate its performance using standard evaluation metrics.
6. Real-Time Implementation: Optimize and deploy the model for real-time emotion recognition in practical environments.

Datasets

Facial: FER2013, CK+, RAF-DB

Speech: RAVDESS, CREMA-D, TESS

Software Requirements

Python, TensorFlow/PyTorch, OpenCV, Librosa, NumPy, Pandas, Flask/FastAPI, Streamlit

Hardware Requirements

Processor (CPU) – Intel i5/i7 or AMD Ryzen 5/7 (8th generation or higher), Graphics Processing Unit (GPU) – NVIDIA GPU with CUDA support (e.g., GTX 1660, RTX 2060 or higher), Memory (RAM) – 16-32 GB, 512 GB SSD, Camera- HD webcam for facial emotion detection, Microphone- High quality for noise cancelling.

Timeline (5 Months)

Week 1-2	March 16-March 29	Understanding the concept, preparation of thesis proposal
Week 3-4	March 30-April 15	Dataset collection and preprocessing
Week 5-6	April 16- April 30	Building CNN model, training on facial data
Week 7-8	May1 -May15	Extract MFCC , train speech

Week 8-9	May16-May 31	Studying fusion methods, implementing and evaluating it
Week10-11	June1-June15	Testing and improving the model
Week 11-12	June16-June30	Real time implementation
Week 13-16	July 1- July 31	Trying for improvements, Final testing and evaluation
Week 17	August1	Thesis report writing

Expected Outcomes

Improved accuracy, real-time system, robust performance

Applications

Healthcare, Education, Driver Monitoring, Customer Service, Human-Computer Interaction.

Conclusion

This proposal develops a multimodal emotion recognition system using facial expressions and speech to overcome limitations of unimodal approaches. By applying advanced models and effective fusion techniques, it aims to achieve high accuracy, robustness, and real-time performance. The system has practical applications in healthcare, HCI, education, and customer service, making it valuable for real-world use.

References

- 1.Jayaswal et al., *"Advances in Facial Expression Recognition Technologies"*, 2025. Shows rapid growth in facial emotion recognition and its importance in computer vision.
- 2.*Computer Science Review*, 2023. Systematic review highlighting the need for multimodal approaches in affective computing
- 3.*Neurocomputing*, 2022. Survey on speech emotion recognition, emphasizing feature extraction and classifier challenges.